

MUHAMMAD WALEED AHMAD

Computer Vision Engineer | Deep Learning | Generative & Agentic AI

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PROFESSIONAL SUMMARY

Results-driven Computer Vision Engineer with 2+ year of hands-on experience building and deploying production-grade computer vision, generative AI, and agentic AI systems. Proven track record delivering real-world products — including a live multi-site surveillance platform (Fortix AI) — using PyTorch, YOLOv11/v26, ByteTrack, OpenCV, LangChain, LangGraph, and RAG architectures. Skilled in end-to-end computer vision MLOps: data annotation, model training and optimization, real-time inference, and deployment with FastAPI and Docker on AWS. Passionate about solving high-impact visual perception problems with intelligent automation.

TECHNICAL SKILLS

Computer Vision: Object Detection & Tracking, Real-Time Inference, Vision Transformers, Multimodal AI, Image Segmentation, 3D Reconstruction, Data Annotation, CVAT, 3D Slicer, FFmpeg, GStreamer, PaddleOCR, EasyOCR, Tesseract OCR, Edge Devices (Nvidia Jetson Orin Nano, Raspberry Pi)

AI / ML: Machine Learning, Deep Learning, NLP, Transformers, LLM Fine-Tuning, Prompt Engineering, Generative AI, Agentic AI, RAG, MLOps, Model Deployment

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Ultralytics YOLO (v11/v26/YOLOE), OpenCV, MediaPipe, LangChain, LangGraph, Hugging Face Transformers, Keras, FastAPI, NumPy, Pandas, N8N

DevOps & Cloud: Docker, AWS, GitHub, CI/CD Pipelines (GitHub Actions)

Languages: Python, JavaScript, Node.js, PostgreSQL, SQLite

WORK EXPERIENCE

Computer Vision Engineer

Shayan Solutions

Oct 2024 – Present

Lahore, Pakistan

- Computer Vision Systems: Designed, trained, and optimized real-time object detection and tracking pipelines using YOLOv11/YOLOv26/SAM3 with ByteTrack, achieving >92% mAP on custom surveillance datasets across 6 detection categories (weapons, fire, PPE, shoplifting, vehicles, faces).
- Agentic AI Pipelines: Built end-to-end multi-agent workflows using LangChain and LangGraph with RAG architectures and vector databases, reducing average query response latency by 40% through optimized retrieval strategies and context window management.
- LLM Fine-Tuning & MLOps: Continuously fine-tuned LLMs using Hugging Face on domain-specific datasets; deployed models via FastAPI and Docker on AWS with GitHub Actions CI/CD, maintaining 99.5% uptime across production systems.
- Scalability & Performance: Engineered feature engineering and data preprocessing pipelines handling 50,000+ video frames per day, improving model inference throughput by 35% through batch optimization and TensorRT integration.

Data Scientist

Fiverivers Technologies

Jun 2024 – Oct 2024

Lahore, Pakistan

- ML Pipelines: Developed end-to-end ML pipelines for 3 client classification and regression projects — covering data collection, cleaning, feature selection, and model evaluation with scikit-learn — improving F1 scores by an average of 18% over baseline models.
- LLM Fine-Tuning & NLP: Fine-tuned 2 LLMs and 1 VLM using Hugging Face on Google Colab and Kaggle (GPU environments); built RAG-based NLP systems with LangChain and LangGraph deployed for real-world client use, processing 10,000+ documents.

PROJECTS

Fortix AI — Real-Time Surveillance System — www.fortixai.io

Stack: YOLOv11/v26/YOLOE, ByteTrack, ANPR, FastAPI, PostgreSQL, AWS

- Built a production web-based multi-site camera monitoring platform serving real clients; implemented real-time detection for 8 event types (fighting, weapons, fire, smoke, PPE violations, shoplifting, robbery, vehicle violations) with <200ms alert latency.
- Integrated facial recognition and Automatic Number Plate Recognition (ANPR) to automatically extract and log license plate text, reducing manual incident reporting time by ~60%.

ANPR System — Automatic Number Plate Recognition

Stack: YOLOv11, PaddleOCR, ByteTrack, FFmpeg, FastAPI

- Built an end-to-end ANPR pipeline using YOLOv11 for license plate detection and ByteTrack for multi-frame vehicle tracking, ensuring each vehicle's plate is recognized once per pass rather than per frame.
- Integrated PaddleOCR for plate text recognition and FFmpeg for RTSP/video stream ingestion and frame extraction, with results served through a FastAPI backend for real-time querying and logging.

Face Swapper Studio — Image & Video Face Swapping

Stack: InsightFace (Inswapper), GFPGAN, RestoreFormer, Python, OpenCV

- Built a face-swapping application for both images and videos, using the InsightFace Inswapper model to accurately map and swap faces onto target subjects.
- Integrated GFPGAN and RestoreFormer enhancement models to restore and sharpen swapped faces, improving output realism and visual quality across varied lighting and pose conditions.

3D Room Reconstruction (Droid-SLAM + Spatial LM)

Stack: Droid-SLAM, Spatial Language Model, Python, 3D Point Cloud Processing

- Implemented Droid-SLAM to generate precise 3D point clouds from monocular room images, achieving <2cm reconstruction error on test scenes; used a spatial LM to reconstruct fully editable 3D layouts for home interior planning use cases.

Advanced Air Drawing — Gesture-Controlled Drawing Application

Stack: MediaPipe, OpenCV, Python, Real-Time Hand Tracking

- Built a real-time computer vision application enabling users to draw on a virtual canvas using hand gestures, powered by MediaPipe hand-landmark tracking for smooth, low-latency interaction.
- Implemented multiple color selection options, dynamic brush thickness adjustment, and gesture-based eraser functionality for an intuitive, touchless drawing experience.
- Added canvas-clear via an open-palm gesture or keyboard shortcut, combining gesture recognition with conventional input handling for a responsive user experience.

EDUCATION

Bachelor of Science in Computer Science

University of Central Punjab

Oct 2021 – Jul 2025

Lahore, Pakistan

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence

ACHIEVEMENTS & HIGHLIGHTS

- Delivered live production AI products, including Fortix AI, within first year of professional career.
- Worked across the full computer vision stack: data collection → annotation → model training → fine-tuning → deployment → monitoring.
- Proficient in both research-oriented (custom model architecture, SLAM) and product-oriented (APIs, MLOps, CI/CD) computer vision engineering.